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CS61C

Great Ideas in Computer Architecture

(a.k.a. Machine Structures)

Datacenters & Cloud Computing

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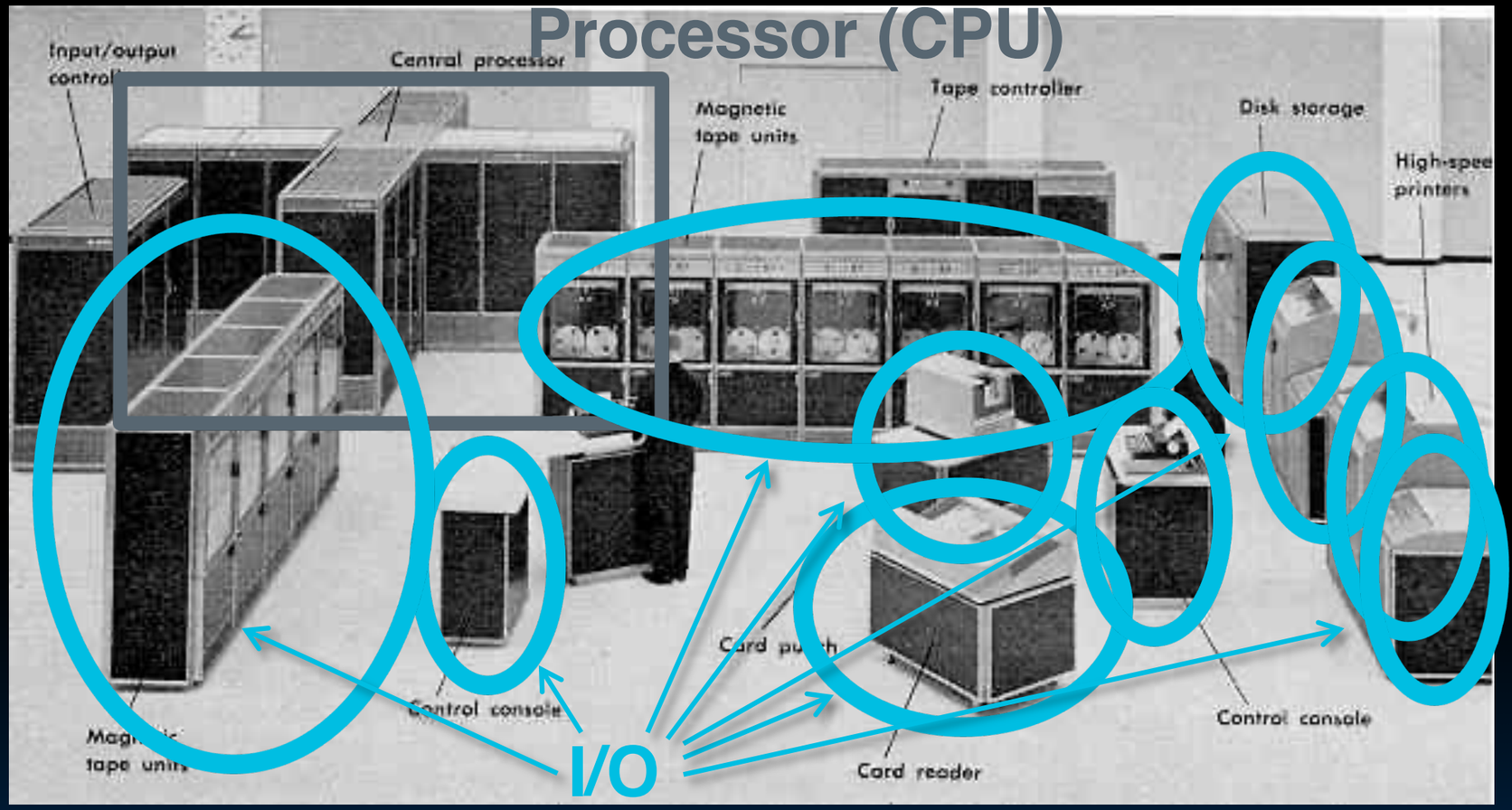
Review … where are we?

- Great Ideas in Computer Architecture
  - ✓ Layers of Representation/Interpretation
  - ✓ Moore's Law
  - ✓ Principle of Locality/Memory Hierarchy
  - Parallelism
  - ✓ Performance Measurement and Improvement
  - Dependability via Redundancy

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Eras of Computer Hardware

Computer Eras: Mainframe 1950s-60s



"Big Iron": IBM, UNIVAC, … build $1M computers for businesses ➔ COBOL, Fortran, timesharing OS

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Minicomputer Eras: 1970s



Using integrated circuits, Digital, HP… build $10k computers for labs, universities ➔ C, UNIX OS

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PC Era: Mid 1980s - Mid 2000s



Using microprocessors, Apple, IBM, … build $1k computer for 1 person ➔ Basic, Java, Windows OS

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PostPC Era: Late 2000s - ??



Personal Mobile Devices (PMD):
Relying on wireless networking, Apple, Nokia, … build $500 smartphone and tablet computers for individuals
➔ Objective C, Swift, Java, Android OS + iOS

Cloud Computing:
Using Local Area Networks, Amazon, Google, … build $200M Warehouse Scale Computers with 100,000 servers for Internet Services for PMDs
➔ MapReduce, Ruby on Rails



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Warehouse Scale Computers



Why Cloud Computing Now?

- "The Web Space Race": Build-out of extremely large datacenters (10,000's of commodity PCs)
  - Build-out driven by growth in demand (more users)
  - Infrastructure software and Operational expertise
- Discovered economy of scale: 5-7x cheaper than provisioning a medium-sized (1000 servers) facility
- More pervasive broadband Internet so can access remote computers efficiently
- Commoditization of HW & SW
  - Standardized software stacks

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November 2024 AWS Instances & Prices

Instance	Per Hour	$ Ratio to Small	Virtual Cores (vCPU)	Memory (GiB)	Disk (GiB)
Standard Small (t3.small)	$0.021	1	2	2	EBS
Standard Large (t3.large)	$0.083	4	2	8	EBS
Standard 2x Extra Large (t3.2xlarge)	$0.333	16	8	32	EBS

- Closest computer in WSC example is Standard 2X Extra Large
- At these low rates, Amazon EC2 makes money! (even utilized 50% time)
- EBS = Elastic Block Store (SSD=$0.08/GB-mo, HDD=$0.045/GB-mo)
- Other options with dedicated attached SSD if you choose & pay for that

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Warehouse Scale Computers

- Massive scale datacenters: 10,000 to 100,000 servers + networks to connect them together
  - Emphasize cost-efficiency
  - Attention to power: distribution and cooling
- (relatively) homogeneous hardware/software
- Offer very large applications (Internet services): search, social networking, video sharing
- Very highly available: < 1 hour down/year
  - Must cope with failures common at scale
- "…WSCs are no less worthy of the expertise of computer systems architects than any other class of machines"

  – Barroso and Hoelzle 2009

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Pioneer of Google's Data Centers Dies at 58 › IEEE also remembers the lives and legacies of other members

BY AMANDA DAVIS | 21 NOV 2023 | 6 MIN READ |



Luiz André Barroso, who was an engineer at Google for more than 20 years, is credited with designing the company's warehouse-size data centers. GOOGLE

Design Goals of a WSC

- Unique to Warehouse-scale
  - Ample parallelism:
    - Batch apps: large number independent data sets with independent processing.
    - Also known as Data-Level Parallelism
  - Scale and its Opportunities/Problems
    - Relatively small number of these make design cost expensive and difficult to amortize
    - But price breaks are possible from purchases of very large numbers of commodity servers
    - Must also prepare for high # of component failures
  - Operational Costs Count:
    - Cost of equipment purchases << cost of ownership

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E.g., Google's Oregon WSC







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Containers in WSCs

Inside WSC

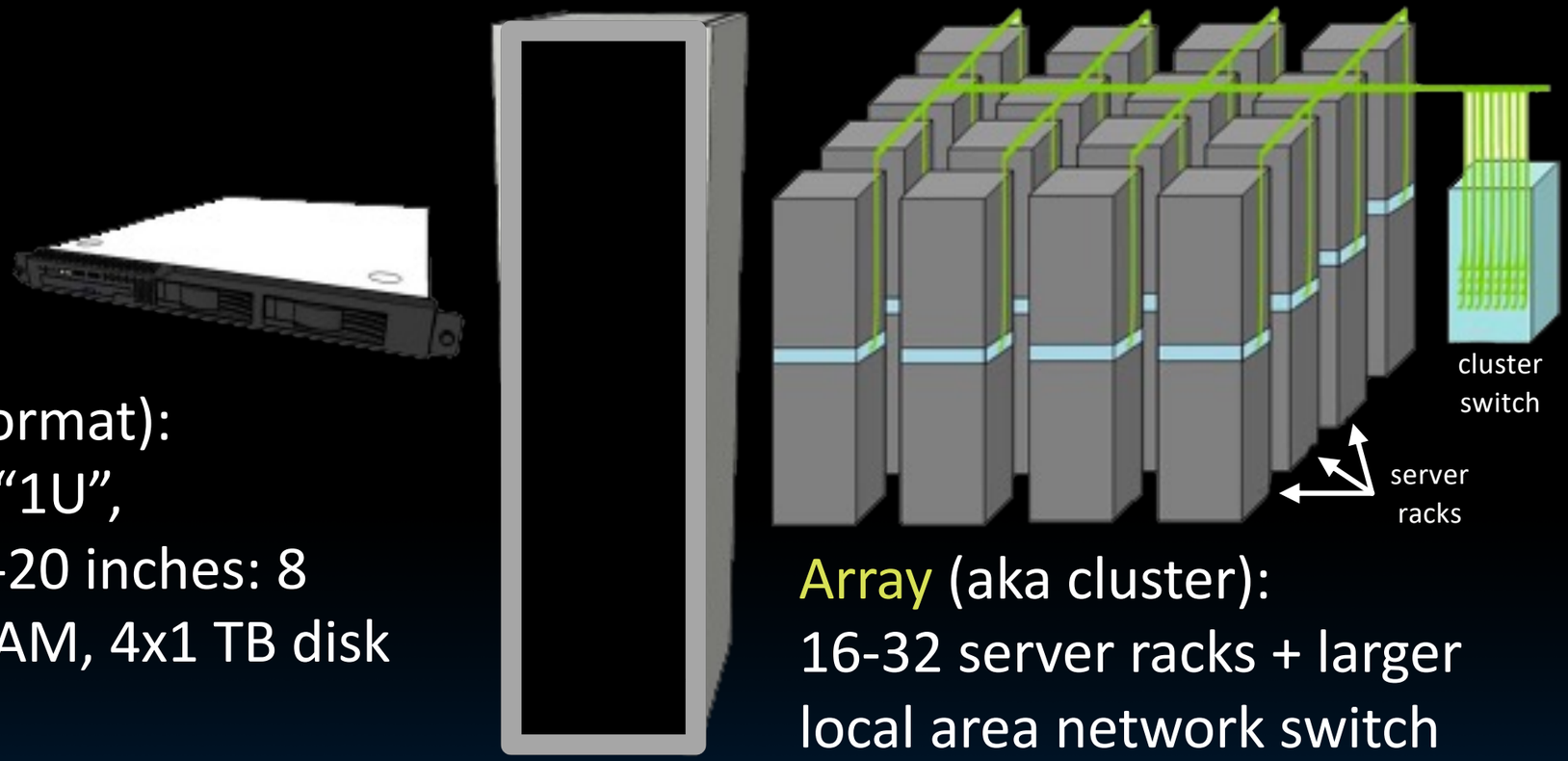


Inside Container



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Equipment Inside a WSC



Server (in rack format):
1 ¾ inches high “1U”,
x 19 inches x 16-20 inches: 8 cores, 16 GB DRAM, 4x1 TB disk

7 foot Rack: 40-80 servers + Ethernet local area network (1-10 Gbps) switch in middle (“rack switch”)

Array (aka cluster):
16-32 server racks + larger local area network switch (“array switch”) 10X faster ➔ cost 100X: cost f(N^2)

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Server, Rack, Array







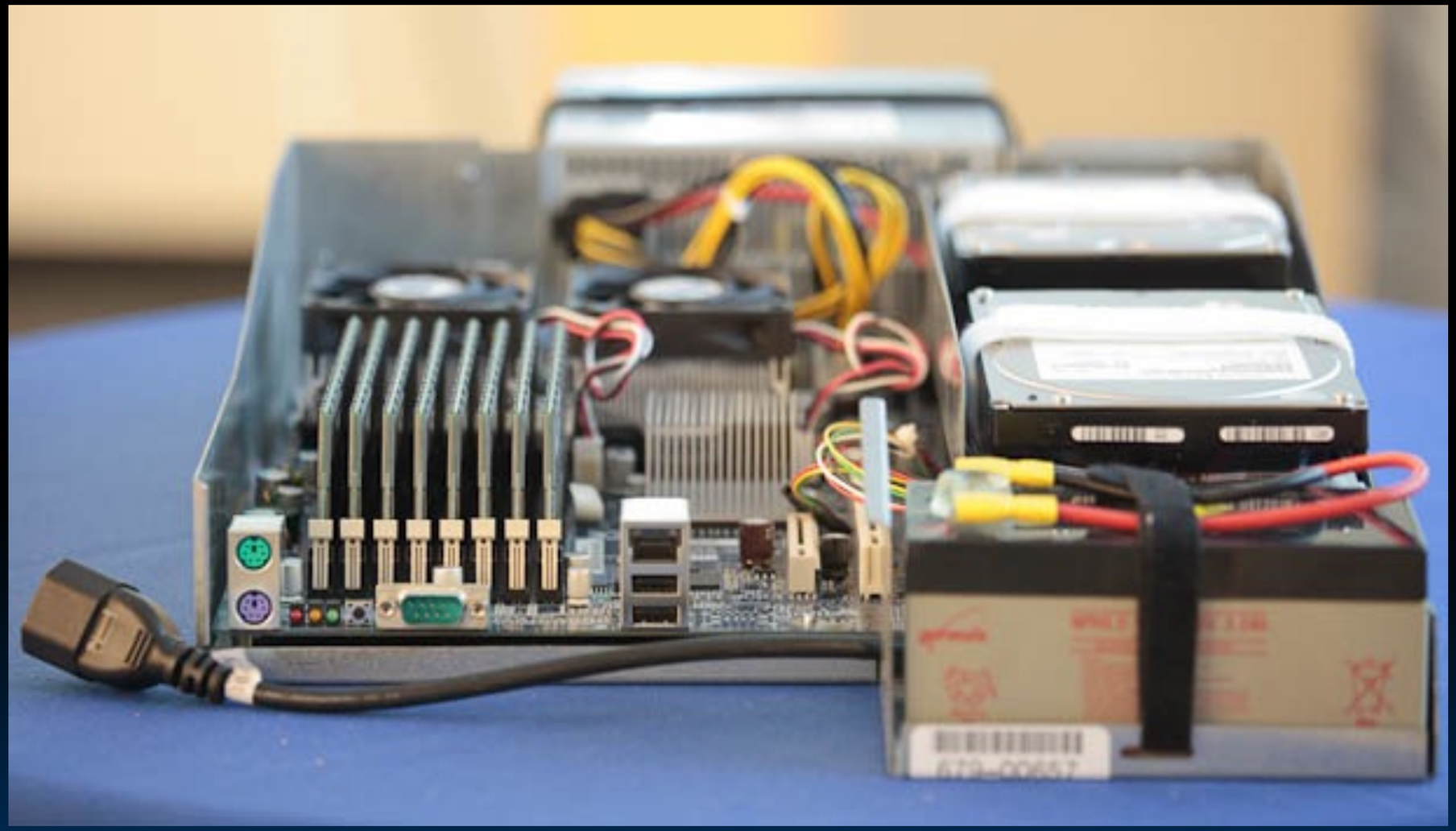
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Google Server Internals



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Google Server Internals



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Defining Performance (review)

- What does it mean to say X is faster than Y?





- 2009 Ferrari 599 GTB
  - 2 passengers, 11.1 secs for quarter mile (call it 10sec)
- 2009 Type D school bus
  - 54 passengers, quarter mile time? (let's guess 1 min)
- Response Time or Latency
  - time between start and completion of a task
  - E.g., time to move vehicle ¼ mile
- Throughput or Bandwidth
  - total amount of work in a given time
  - E.g., passenger-miles in 1 hour

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Coping with Performance in Array

Lower latency to DRAM in another server than local disk

Higher bandwidth to local disk than to DRAM in another server

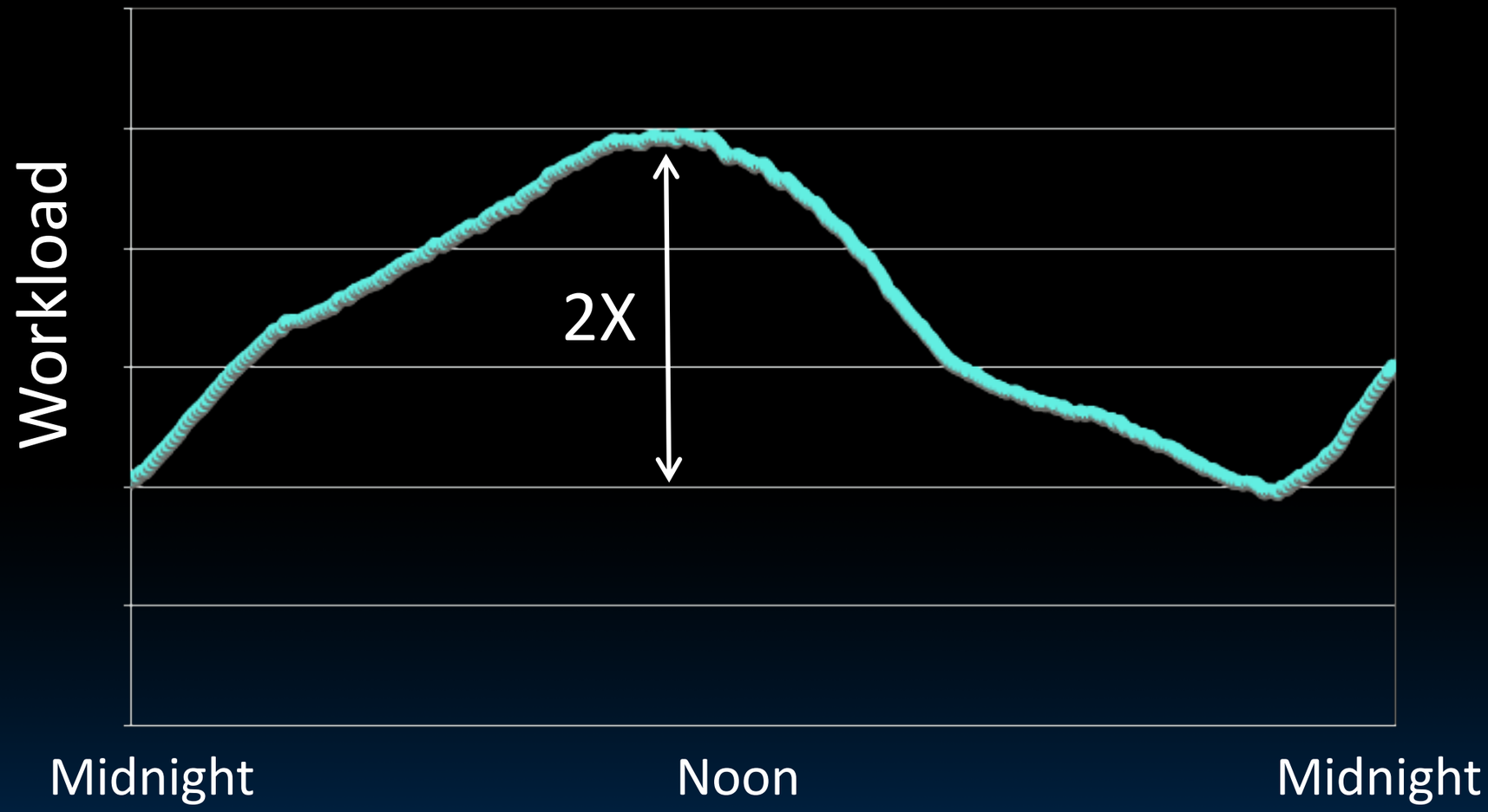
	Local	Rack	Array
Racks	--	1	30
Servers	1	80	2400
Cores (Processors)	8	640	19,200
DRAM Capacity (GB)	16	1,280	38,400
Disk Capacity (TB)	4	320	9,600
DRAM Latency (microseconds)	0.1	100	300
Disk Latency (microseconds)	10,000	11,000	12,000
DRAM Bandwidth (MB/sec)	20,000	100	10
Disk Bandwidth (MB/sec)	200	100	10



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Power Usage Effectiveness (PUE)

Coping with Workload Variation



- Online service: Peak usage 2X off-peak

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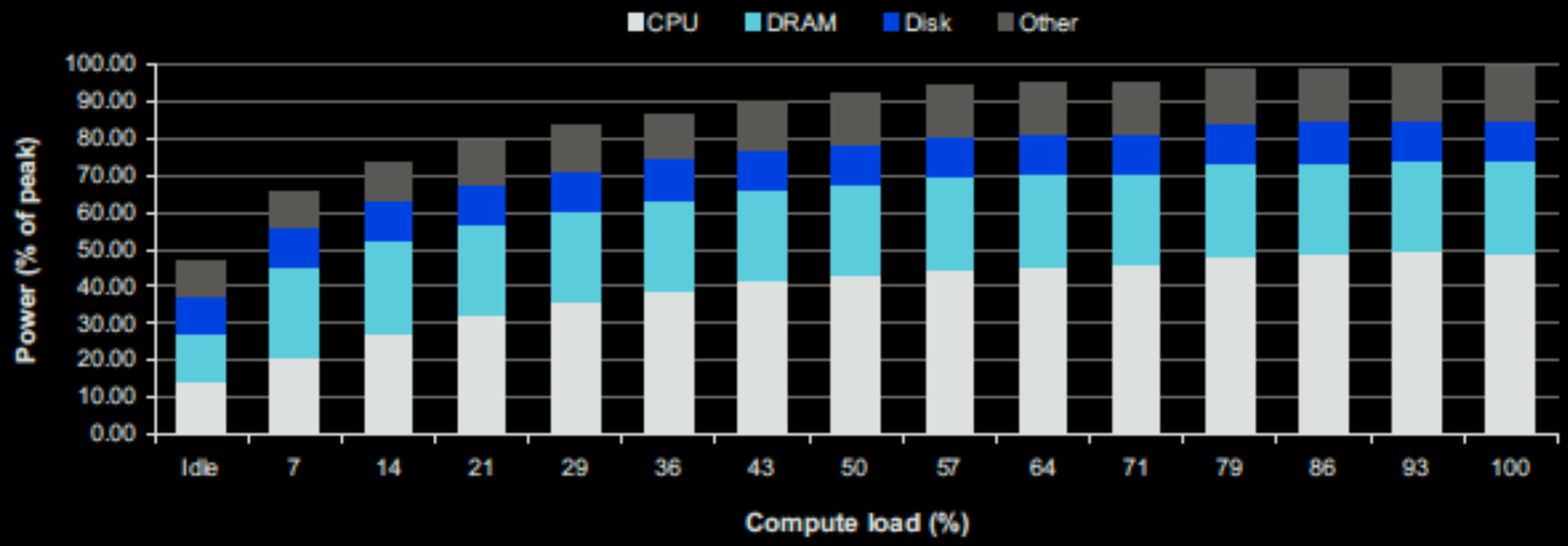


Impact of latency, bandwidth, failure, varying workload on WSC software?

- WSC Software must take care where it places data within an array to get good performance
- WSC Software must cope with failures gracefully
- WSC Software must scale up and down gracefully in response to varying demand
- More elaborate hierarchy of memories, failure tolerance, workload accommodation makes WSC software development more challenging than software for single computer

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Power vs. Server Utilization



- Server power usage as load varies idle to 100%
- Uses ½ peak power when idle!
- Uses ⅔ peak power when 10% utilized! 90%@ 50%!
- Most servers in WSC utilized 10% to 50%
- Goal should be Energy-Proportionality:
  % peak load = % peak energy

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Power Usage Effectiveness

- Overall WSC Energy Efficiency: amount of computational work performed divided by the total energy used in the process
- Power Usage Effectiveness (PUE):
  Total building power / IT equipment power
  - A power efficiency measure for WSC, not including efficiency of servers, networking gear
  - 1.0 = perfection

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PUE in the Wild (2007)

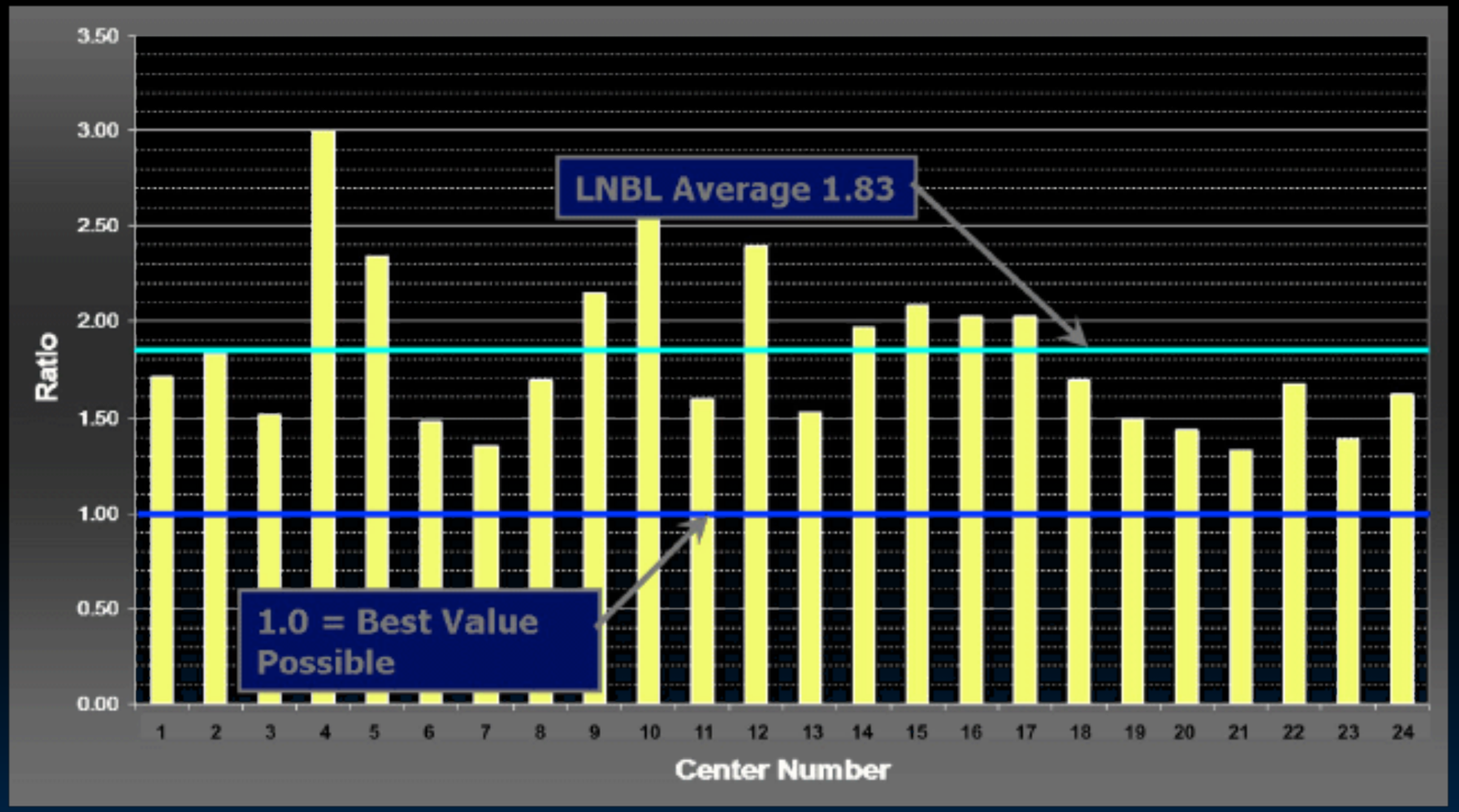


FIGURE 5.1: LBNL survey of the power usage efficiency of 24 datacenters, 2007 (Greenberg et al.)

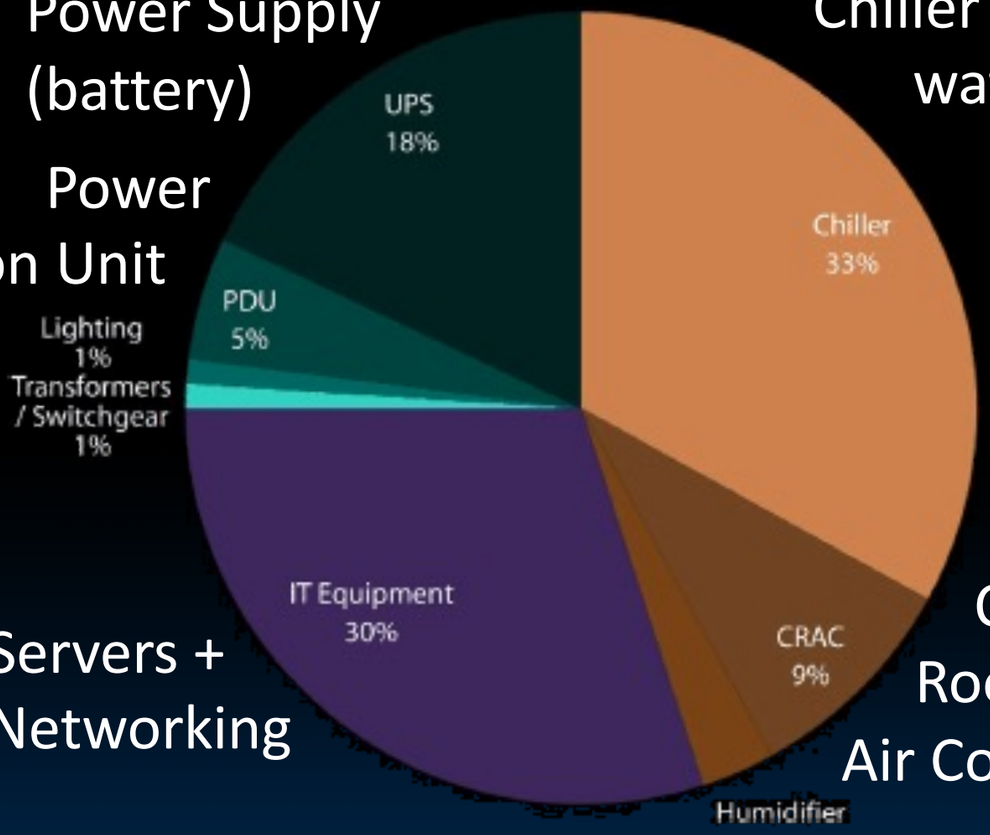
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High PUE: Where Does Power Go?

Uninterruptable Power Supply (battery)

Power Distribution Unit

Chiller cools warm water from Air Conditioner



Servers + Networking

Computer Room Air Conditioner

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Google WSC A PUE: 1.24

- Careful air flow handling
  - Don't mix server hot air exhaust w/cold air (separate warm aisle from cold aisle)
  - Short path to cooling so little energy spent moving cold or hot air long distances
  - Keeping servers inside containers helps control air flow
- Elevated cold aisle temperatures
  - 81º F instead of traditional 65º - 68º F
  - Found reliability OK if run servers hotter
- Use of free cooling
  - Cool warm water outside by evaporation in cooling towers
  - Locate WSC in moderate climate so not too hot or too cold
- Per-server 12-V DC UPS
  - Rather than WSC wide UPS, place single battery per server board
  - Increases WSC efficiency from 90% to 99%
- Measure vs. estimate PUE, publish PUE, and improve operation

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Computing in the News



Urs Hölzle, Google SVP
Co-author of today's reading

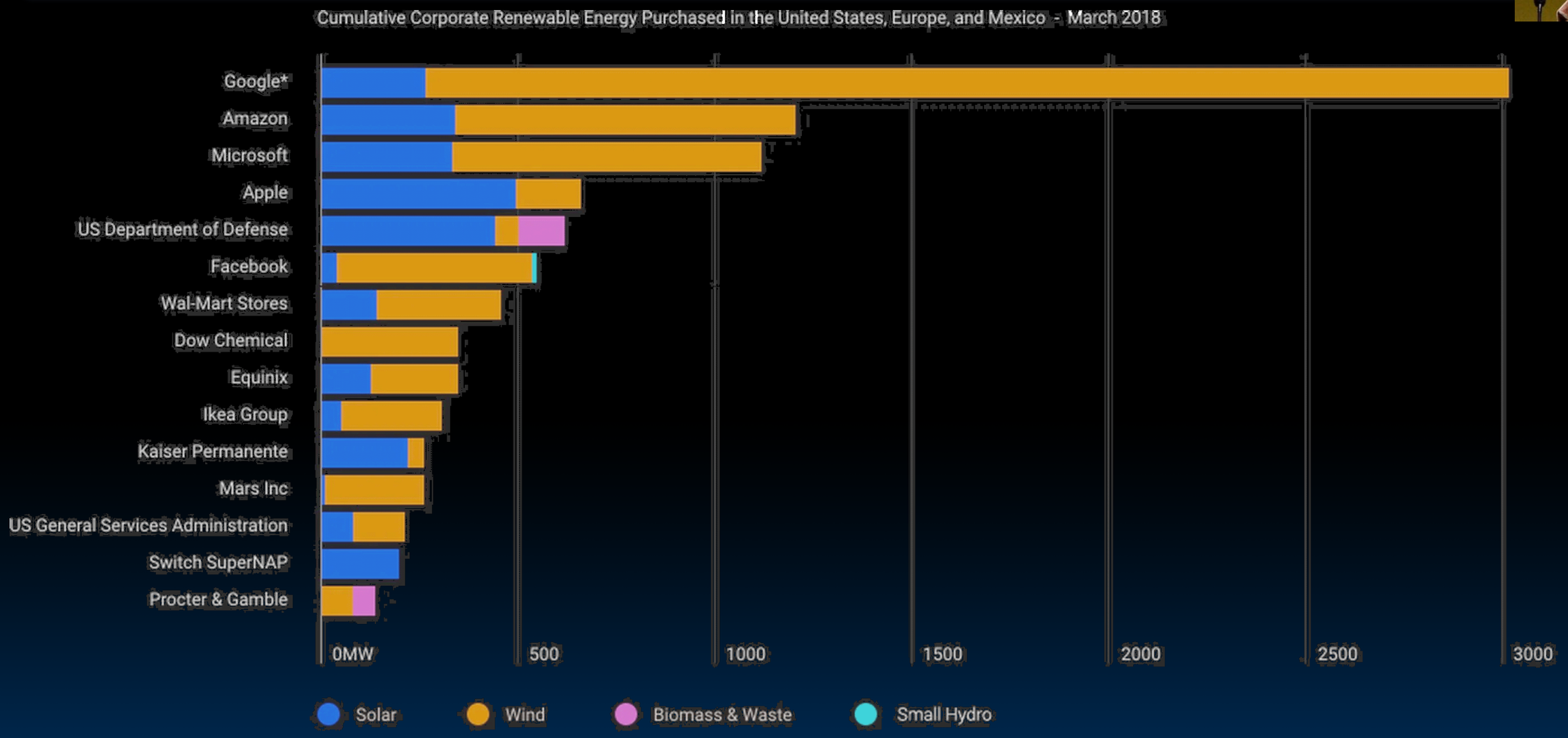
- 2011 www.nytimes.com/2011/09/09/technology/google-details-and-defends-its-use-of-electricity.html
  - Google disclosed that it continuously uses enough electricity to power 200,000 homes, but it says that in doing so, it also makes the planet greener.
  - Search cost per day (per person) same as running a 60-watt bulb for 3 hours
- 2018 techcrunch.com/2018/04/04/google-matches-100-percent-of-its-power-consumption-with-renewables/
  - Google: "Over the course of 2017, across the globe, for every kilowatt-hour of electricity we consumed, we purchased a kilowatt-hour of renewable energy from a wind or solar farm that was built specifically for Google. This makes us the first public Cloud, and company of our size, to have achieved this feat"

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Computing in the News

Urs Hölzle, Google SVP
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Summary

- Parallelism is one of the Great Ideas
  - Applies at many levels of the system – from instructions to warehouse scale computers
- Post PC Era:Parallel processing, smart phone ↔ WSC
- WSC SW must cope with failures, varying load, varying HW latency bandwidth
- WSC HW sensitive to cost, energy efficiency
- WSCs support many of the applications we have come to depend on



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